

Autonomous Scientific Director and Research-Operations Architecture

Status: METHODS / RESEARCH-OPERATIONS CASE — INTERNAL CHECKPOINT; EXTERNAL VALIDATION PENDING

Research question

Can a human-authorized, AI-assisted research operation select among heterogeneous scientific targets, produce independently checkable work, preserve corrections and negative results, and stage publication-ready packages without crossing human-only publication, identity, legal, or public-release boundaries? The architecture treats research quality as a control problem spanning target selection, evidence, verification, reproducibility, privacy, and handoff.

Contribution and current result

The versioned director workspace records a completed 3/3 core objective across three distinct campaigns: Graffiti3 finite-group refutations; semiorde zonotope bounds; and the Erdős #64 excess-degree theorem package. Each package has a separate claim ledger, manuscript/supplement, reproducibility archive, checksums, verification gates, and private staging record. The latest checkpoint marks the three campaigns as scientifically distinct rather than an artificial split and selected an Erdős #409 F=104 replay as the next target without starting it after the deadline.

The system also preserved protected states: RH remained unsolved/in progress; Study 14's confirmatory core and held-out outcomes remained unchanged; no email, journal submission, preprint submission, public release, repository-visibility change, or legal/authorship attestation was performed.

Methods

The workflow combines portfolio-level target comparison, frozen campaign contracts, source registries, append-only state, proof reconstruction from definitions, exact computation, method-distinct replay, hostile audit, mutation testing, prior-art scoping, PDF inspection, checksum and zero-mutation archive checks, correction/supersession records, private staging, and versioned handoff bundles. Heavy computation and model-worker concurrency are bounded. Scientific validity, novelty, external review, publication readiness, and public authorization are tracked separately.

Zackary's role

Zackary is the human program owner and authorization boundary. He set the portfolio objective and constraints, supplied the research corpus, and retained sole control over identity, authorship, affiliation, conflicts, funding, licensing, submission, email, and public visibility. Recorded Study 14 episodes show one initial authorization and no post-freeze human route selection; the system's autonomous choices therefore remain distinguishable from Zackary's final legal and publication decisions.

AI and tool role

Codex and other AI systems performed bounded discovery, target selection, mathematical exploration, code generation, proof reconstruction, adversarial checking, documentation, and reversible private staging. Local tools included Git, Python, GAP, C, exact data tables, LaTeX/PDF tooling, hashes, and structured YAML/JSON ledgers. Multiple AI lanes provide internal cross-checking only; they are not external peer review.

Verification

The final checkpoint is frozen at commit d959ec7db9c1910be01623e0b0de0b00042d1599 with checksummed Git bundle and portable archive. The three packages report 15/15, 21/21, or campaign-specific passing gates and exact replay records. Those checks validate their declared scopes; the methods architecture itself has not been compared in a randomized study, audited by an external research group, or shown to improve outcomes causally.

Exact nonclaims

The workflow is not claimed to be autonomous science without human responsibility, a replacement for peer review, or proof that ETI outperforms baselines. Three prepared packages are not three published or peer-reviewed papers. Naturalistic episodes are not causal efficacy evidence. No global open problem is claimed solved, and private staging is not publication.

Public artifacts

No public director checkpoint yet. Supporting public projects, which do not expose the full director architecture, include [Erdős #64](#) and [Erdős #409](#).

Employer relevance

The case demonstrates agent orchestration, research program management, verification pipelines, data lineage, correction handling, privacy controls, reproducible packaging, and responsible escalation to a human decision-maker. It is relevant to AI research operations, research engineering, formal verification, technical program management, and safety-critical knowledge workflows.